

## Review

# Use of IoT sensor devices for efficient management of healthcare systems: a review

G. Gopichand<sup>1</sup> · T. Sarath<sup>2</sup> · Ankur Dumka<sup>3</sup> · Himanshu Rai Goyal<sup>4</sup> · Rajesh Singh<sup>5</sup> · Anita Gehlot<sup>5</sup> · Lovi Raj Gupta<sup>6</sup> · Amit Kumar Thakur<sup>6</sup> · Neeraj Priyadarshi<sup>7</sup> · Bhekisipho Twala<sup>8</sup>

Received: 14 May 2024 / Accepted: 2 August 2024

Published online: 05 August 2024

© The Author(s) 2024 [OPEN](#)

## Abstract

With an aging population and an increase in chronic sickness, healthcare systems are under increasing pressure. Internet of Things (IoT) technology has recently garnered a lot of interest for its ability to alleviate this burden. The IoT is a network of interconnected wireless digital devices that can gather, transmit, and store data automatically, without the need for human or computer intervention. In order to better anticipate health problems, diagnose, treat, and monitor patients in and out of the hospital, the IoT offers numerous advantages that could streamline and improve health care delivery. As a matter of urgency, decision-makers and heads of state around the world are enacting policies to provide healthcare services through the use of technology. Thanks to recent advancements in the IoT, healthcare has been able to evolve at a rapid pace. Research on Internet IoT uses in healthcare and newly developed emergency equipment are summarized in this article.

**Keywords** Internet of Things · Healthcare · Sensor devices · Blockchain · Patient care · Monitoring · Patients records

## 1 Introduction

Medical treatment is fundamental to human survival. Hospital beds, doctors, and nurses are in great demand, and the demand for healthcare services is skyrocketing due to the ageing population and the accompanying increase in chronic illness [1]. Clearly, there must be a fix that will alleviate the strain on healthcare systems without sacrificing the quality of treatment that give to people who are most vulnerable.

Much recent research has focused on the Internet of Things (IoT) since it has been accepted as a viable solution to reduce the demands on healthcare systems [2, 3]. The IoT industry has seen an influx of researchers in the past few years [4, 5]. For example, by 2025, the industrial Internet of Things market is expected to reach 110.6 billion USD, according to Markets. In addition, by 2030, experts anticipate that there will be a massive network of interconnected devices, including everything from smartphones to home appliances, with roughly 50 billion IoT items and devices in use around the globe.

---

✉ Amit Kumar Thakur, amittthakur3177@gmail.com | <sup>1</sup>School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India. <sup>2</sup>School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore, India. <sup>3</sup>Department of Computer Science, Women Institute of Technology, VMSBUTU, Dehradun, Uttarakhand, India. <sup>4</sup>Department of CSE, Graphic Era Deemed to Be University, Dehradun, Uttarakhand, India. <sup>5</sup>Uttaranchal Institute of Technology, Uttaranchal University, Dehradun 248007, India. <sup>6</sup>Lovely Professional University, Phagwara, Jalandhar, Punjab 144001, India. <sup>7</sup>Department of Electrical Engineering, JIS College of Engineering, Kolkata 741235, India. <sup>8</sup>Digital Transformation Portfolio, Tshwane University of Technology, Staatsartillerie Rd, Pretoria West, Pretoria 0183, South Africa.



By allowing for the early diagnosis and prevention of diseases and other potentially harmful conditions, the IoT implementation in remote health monitoring offers tremendous potential to improve the quality of healthcare services while reducing costs [6, 7]. There are several potential medical applications for the IoT, including the management of chronic illnesses, care for the elderly, and fitness activities. Therefore, a possible answer is to employ technology that can remotely monitor sick people. The IoT has the potential to reduce healthcare expenditures and improve patient outcomes by collecting and transferring real-time health data from patients to clinicians [6]. One essential component of the Internet of Things is smart devices, and one example is medical imaging and diagnostic equipment, which includes sensors. Furthermore, it is expected that medical services built on IoT would increase efficiency, prolong patient lives, and provide several benefits to its customers. Also, more patients can get high-quality care thanks to IoT's ability to optimize the allocation of limited resources. They also lend a hand when it comes to medical server design, gateway integration, and the provision of on-demand healthcare to authorized parties.

So, this paper's original contribution is that it catalogues every essential part of an IoT healthcare system from beginning to finish, with an emphasis on sensors that track different health metrics, standards for both short- and long-range communications, and cloud computing. The contributions taking into account all the necessary parts of healthcare systems that rely on the Internet of Things.

## 2 Architecture of IoT based healthcare

The three primary components of the Internet of Things (IoT) in healthcare delivery are the following: (1) the perception layer; (2) the network layer; and (3) the application layer [8].

### 2.1 Layer of perception: data-collecting sensing systems

The Internet of Things is built around technologies that enable perception and identification. Wireless Identification, Infrared (IR) Sensors, Cameras, Global Positioning Systems (GPS), medical devices, and smart devices are all examples of sensors. These sensors can convert information to digital signals, making it easier to transmit via a network, and they provide comprehensive perception through object recognition, location recognition, and geography recognition [8, 9]. Unfortunately, not all IoT sensor devices have been clinically validated or proven to be safe or effective, even though there are many examples of gadgets that could possibly save lives.

### 2.2 Layer of the network: information transfer and data storage

Both wired and wireless networks make up the network level of the Internet of Things (IoT), which is responsible for communication and the local or central storage of processed (layer 1) information. While low, medium, and high frequencies are all viable options for inter-thing communication, the Internet of Things is primarily concerned with the latter. Some examples of such technologies are Radio Frequency Identification (RFID), Bluetooth, Zigbee, low-power Wi-Fi and the global system for mobile communications [8].

Data transmissions can be made to a central server in the cloud or kept locally, which is frequently a decentralized approach. The widespread availability, adaptability, and scalability of cloud computing make it an ideal platform for supporting the delivery of health services via data capture, storage, and transfer [47]. Electronic Medical Records (EMRs), patient portals, medical IoT devices (such as smartphone apps), and the big data analytics powering decision support systems and treatment methods are all likely to make use of the cloud in the near future [10].

The IoT might be more scalable in healthcare if data processing and networking were decentralized. One relatively recent idea in cloud computing is the "edge cloud," which enables IoT sensors and network gateways to decentrally handle and analyze data at the edge, rather than transmitting and managing it all from a central place [8, 10]. Blockchain storage is similar in that it employs a decentralized method of data storage by constructing separate blocks that hold distinct sets of information; each block then becomes an interdependent link in a larger block, forming a network that is controlled by patients themselves, as opposed to an outside entity [11].

## 2.3 Layer for application

The application layer is in charge of providing the user with services that are specific to the application, as well as interpreting and applying data. When it comes to the Internet of Things (IoT), Artificial Intelligence (AI) has some very exciting potential medical uses. AI has several scientific uses, such as image analysis, pharmacological activity design, text recognition with natural language processing, and gene mutation expression prediction [12]. In order to make treatment and/or diagnosis decisions, AI may interpret and contextualise electronic medical record data, which includes medical history, physical, laboratory, imaging, and prescriptions.

By combining the IoT with deep machine learning, healthcare providers will be able to see what they can't see before, leading to improved diagnostic capabilities. While complete certainty in diagnoses is unlikely to be achieved, the integration of machine learning with human clinical knowledge consistently improves system efficiency. This is seen, for instance, in the following areas: mental health and diabetes [13], congestive heart failure [14, 15], bone disease [16], Alzheimer's disease [17], classification of benign and malignant tumors [18, 19], and cardiac arrhythmias [20].

## 3 Improvement of health service delivery with IoT

The burden of disease related to modifiable risk factors is higher than ever before [21, 22], thus disease prevention must become a priority this decade. A genuine hybrid model of primary, secondary, and tertiary care could be possible with the help of the IoT in health care, which could lead to better population health and allow the health system to make better use of its current personnel.

### 3.1 Expanding access to primary healthcare

More than 90% of lifestyle self-management occurs outside of hospitals and clinical settings, by patients themselves, even among high health care consumers [23, 24], making this transformation of health delivery critical for increasing self-management for individuals with chronic diseases. The need for easily accessible health information is evident in the public demand. Take a 2015 US survey as an example; 58% of smartphone users (931/1604) used a health-related app to self-manage their lifestyle [25]. AI has also accelerated the dissemination of health information at the point of care, with chatbots (also known as AI physicians) being able to provide medical and lifestyle recommendations. There is a lack of a formal procedure for approving applications or informing customer choice, and there is still a lot of work to be done to understand how chatbots can enhance health. Additionally, over 50% of the top-rated apps make unapproved medical claims [26]. Consequently, it is crucial to have a solid evidence base for digital health [27]. Digital prescriptions have the potential to promote the use of the IoT in healthcare and encourage a broader public emphasis on illness prevention if medical practitioners have easy access to evidence-based digital resources, gadgets, and smartphone applications.

### 3.2 Integrative, ongoing, and preventative health care at the secondary and tertiary levels

By leveraging the Internet of Things (IoT), health care systems can improve their service delivery model from the current reactive, intermittent, and disjointed state to one that is proactive, continuous, and coordinated [28]. The potential for less intrusive, high-quality treatment that is attractive to both patients and medical staff is a major selling point for this method. Policymakers are very interested in this shift in the health care system's landscape because it has the potential to drastically improve the system's efficiency (and thus decrease resource use) [29] and give the system the leeway to adapt its care models and service delivery to meet the needs of individuals or populations.

Doctors were previously unable to use medical gadgets that could do real-time analysis due to recent advancements in IoT technology. It has also helped healthcare facilities serve more patients at once and keep costs down while maintaining high-quality treatment. As a result, the patient was able to participate more actively in their treatment while experiencing less financial strain. Disease diagnosis, personal care for pediatric and elderly patients, health and fitness management, and supervision of chronic diseases are some of the HIoT applications that have evolved thanks to the significant impact of the IoT in recent years. It has been split into two main groups, services and apps, to help you understand these uses better. Concepts utilized in the development of HIoT devices are part of the former, while healthcare applications in the diagnosis of certain health conditions or measures of health parameters are part of the latter. Here we will go over the services and applications of the HIoT in great depth.

## 4 Concepts and services in HIoT

The introduction of new services and ideas has revolutionized healthcare by addressing long-standing issues. Due to increasing healthcare demands and technological advancements, more and more services are being added on a daily basis. The incorporation of these into the design of an HIoT system is rapidly expanding. In a healthcare-oriented IoT setting, each service offers a variety of options. These concepts and services are not defined in a unique way. The uses for which HIoT systems are implemented are what set them apart. But to help you get a feel for the subject, we've included a graphic representation of some of the most popular healthcare IoT services in the Fig. 1

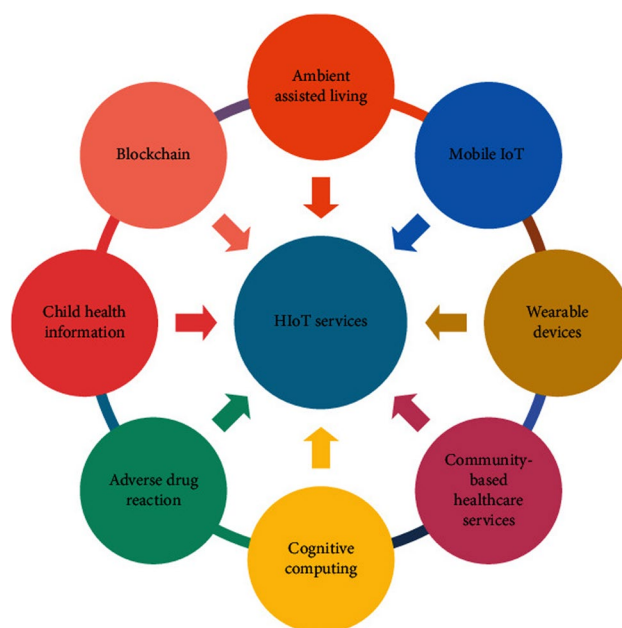
### 4.1 Ambient assisted living

A subfield of AI known as "Ambient Assisted Living" (AAL) uses the Internet of Things (IoT) to aid the elderly. One of the primary goals of AAL is to make it easier and safer for the elderly to continue living independently in their own homes. AAL offers a method for keeping tabs on these patients in real time and ensuring that they will receive support similar to human services in the event of a medical emergency. When applied to the healthcare industry, cutting-edge AI technologies like big data analysis and machine learning make this a reality. Researchers have mostly focused on three primary areas of activity-based AAL: activity recognition, environment recognition, and vital monitoring. The most buzz, though, was around activity recognition systems, since they help identify dangerous situations or urgent medical issues that could compromise the health of the elderly. Figure 2 depicts the fundamental design of an AAL smart healthcare framework. A large body of research has documented the use of the IoT in AAL [21–24].

In general process of AAL, first step is Health Monitoring which is a Vital signs and other health parameters are constantly checked to find possible health problems early. For example, a wearable device that checks your heart rate, blood pressure, and sleep patterns and lets carers and healthcare workers know if something is wrong. The second step is emergency response function finds and helps with emergencies right away, like falls or medical situations. An example of a fall detection device is one that lets emergency contacts or services know when someone falls. The third step is Daily Living Assistance: Help with daily tasks like taking medications, making meals, and doing jobs around the house. For example, a smart pill dispenser that tells carers if a dose is missed and remembers the user to take their medicine.

the final step is Social Connectivity Function: Help older people make friends and feel less alone by improving their social connections. For example, a video talking system that is built into a TV makes it easy to talk to family and friends. In this way the AAL will generate data for analysis.

Fig. 1 Services in HIoT



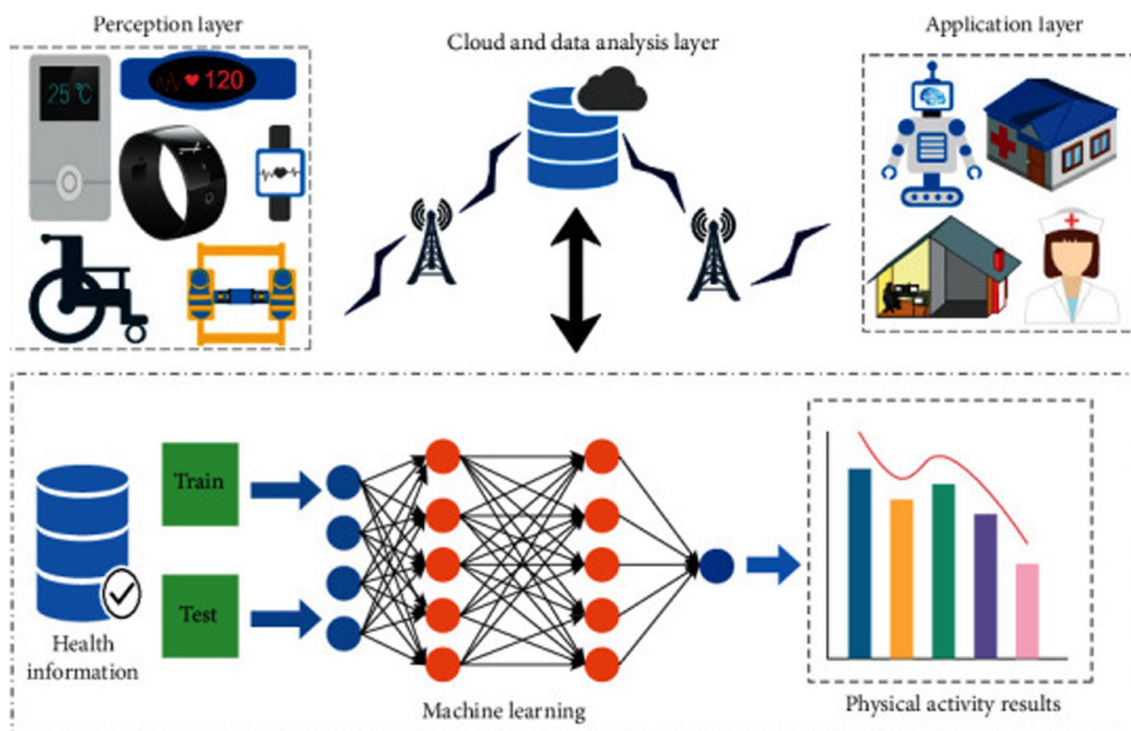


Fig. 2 Fundamental Design of AAL (Google Courtesy)

## 4.2 Mobile IoT

The term “Mobile IoT” (or “m-IoT”) describes the network of interconnected computing devices, sensors, communication networks, and cloud services used to monitor vital signs and other health data collected from patients (Fig. 3). Thus, it facilitates an effective Internet-based healthcare service by acting as a communication bridge between

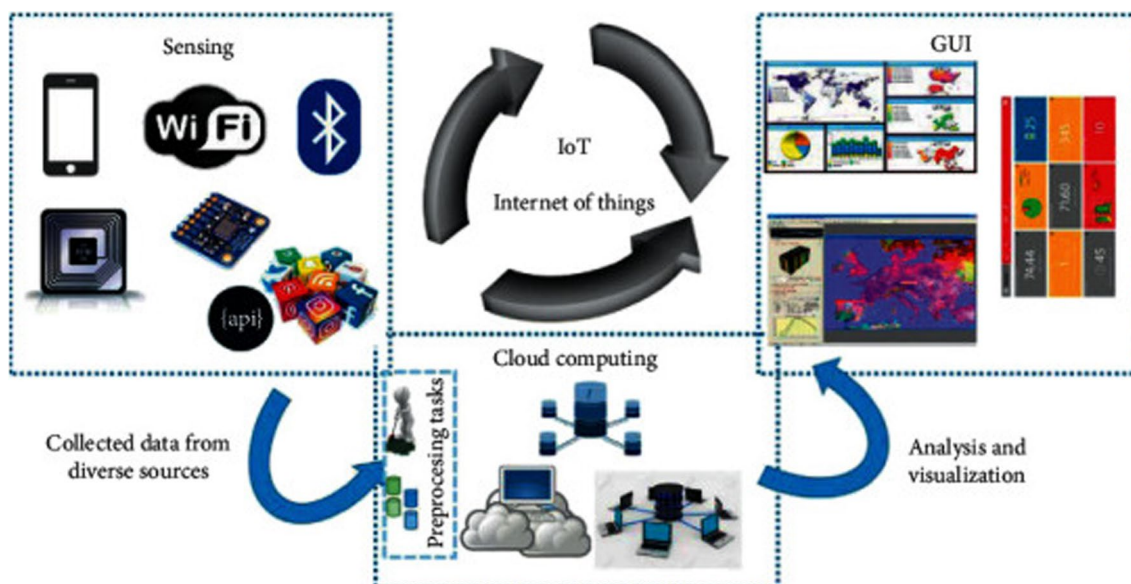


Fig. 3 Mobile IoT Working (Google Courtesy)



mobile networks such as 4G and 5G and personal area networks [25]. Healthcare providers now have easier access to patient data, faster diagnosis, and more efficient treatment thanks to mobile HIoT services.

### 4.3 Wearable gadgets

When it comes to managing a wide range of health concerns, wearable gadgets assist both patients and healthcare providers in a cost-effective manner. Various sensors can be integrated with common wearable accessories used by people, such as watches, wristbands, necklaces, shirts, shoes, handbags, caps, and so on as shown in Fig. 4, to create these non-invasive devices [26, 27]. Data on the patient's health and the surrounding environment can be collected by means of the attached sensor. After that, the data is sent to the server or database.

### 4.4 Healthcare network that encompasses a local community

Creating a healthcare network that encompasses a local community like a private clinic, a small residential area, a hotel, etc. to monitor the health conditions of the individuals live there is the idea behind community-based healthcare monitoring. A community-based network is one in which multiple networks are joined together to provide a service that allows users to work together. For the purpose of healthcare monitoring in outlying regions, an IoT-based cooperative medical network was established in [28]. Various authentication and authorization procedures were utilized to establish a secure connection between the networks.

### 4.5 Cognitive approach

Computing using a cognitive approach mimics the way the human brain processes information. The Internet of Things (IoT) has evolved to include sensors that can solve issues in a way similar to the human brain, thanks to developments in AI and sensor technology. When integrated into an IoT system, cognitive computing facilitates the discovery of previously unseen patterns within massive datasets [29]. In addition, it improves a sensor's capability to autonomously adapt to its environment and process healthcare data. Every sensor in a cognitive IoT network works in tandem with every other smart device to deliver effective healthcare. Integrating cognitive computing into an IoT system allows medical



Fig. 4 Wearable Gadgets (Google Courtesy)

professionals to better monitor patient data and administer appropriate treatment. A smart healthcare monitoring system that employs EEG and cognitive computing to determine the patient's pathological condition has been proposed in [30]. In order to determine the patient's health, the EEG data was combined with data from other sensors that collected information about the patient's speech, gestures, bodily movements, and facial expressions. It also makes it easier to get immediate medical assistance in the event of a dangerous situation.

#### 4.6 Adverse drug reactions

Medication side effects are known as Adverse Drug Reactions (ADRs). A single dose or repeated treatment over time could trigger the response. Because of the potential for an unfavorable reaction while taking two medications simultaneously, this is also something that could happen. ADRs can occur in any given patient and are not specific to any one illness or medication. At the patient's terminal, an IoT-based ADR system uses a distinct identity or barcode to identify each medication [31]. A pharmaceutical intelligent information system can verify the drug's compatibility with the patient's body. Each patient's allergy profile is saved in the system utilizing electronic health records.

#### 4.7 Blockchain technologies

An essential component of any HIoT network is the exchange of data between various healthcare providers and medical equipment. Data fragmentation, however, is a big problem when it comes to safe data exchange. When data is fragmented, it could cause healthcare providers linked to a single patient to have incomplete or incorrect information. The therapy process could be hindered by a lack of knowledge. Healthcare facilities can connect all of the data repositories in their network with the use of blockchain technology, which eliminates data fragmentation [32]. This improves communication between healthcare providers and their patients and further guarantees the safe transfer of personal health information. Figure 5 shows that blockchain technology also encourages healthcare practitioners and organizations to work together to do qualitative research. There are three possible causes for blockchain technology's secure transmission. One important feature is its immutable "ledger" that users can view and modify. It guarantees that the ledger records cannot be altered once added. On top of that, there are rules that all ledger transactions must adhere to. Second, blockchain may run on numerous computers, devices, etc., all at once because it is a distributed ledger. The third feature is that blockchain technology uses a smart contract mechanism to adhere to the rules of agreements and policies for data exchange. The smart contract controls who can access which blockchain-stored electronic medical records (EMRs) and how. What this means is that doctors can only access the electronic medical records (EMRs) that they have been granted access to. For electronic medical record (EMR), medication prescription, and clinical pathway management, the healthcare industry has seen a proliferation of blockchain projects in the last several years [33–35].

IoT healthcare systems can benefit greatly from blockchain technology because it improves data security, interoperability, patient privacy, and the ease of data management. It is a strong tool for solving some of the biggest problems in healthcare because it can provide records that can't be changed, are spread out, and are open to everyone. Blockchain can help the healthcare industry improve patient outcomes, streamline operations, and encourage new ideas. This will create a safer, more efficient, and patient-centered healthcare environment in the long run.

#### 4.8 Child health information

The idea of Child Health Information (CHI) centres on raising consciousness about the importance of a kid's health. The primary goal of CHI is to provide parents and children with the knowledge and tools necessary to make informed decisions about a child's physical, mental, and emotional well-being. Researchers have developed a platform that can monitor and regulate a child's health, thanks to the application of IoT. An IoT framework that allows for the monitoring of a child's emotional and physiological well-being [36]. In addition, parents and medical professionals can work together to take the required steps in the event of an emergency. An Internet of Things (IoT) medical network that links a medical device to a smartphone app was created in a related study [37]. The technique measures five distinct physiological variables: stature, core temperature, systolic blood pressure, and weight. This data is accessible to medical practitioners and other health care providers through the app. In order for parents and teachers to keep tabs on their children's eating habits, an m-health service has been suggested in [38]. The kids were able to get the healthy nutrients they needed with the help of the app.

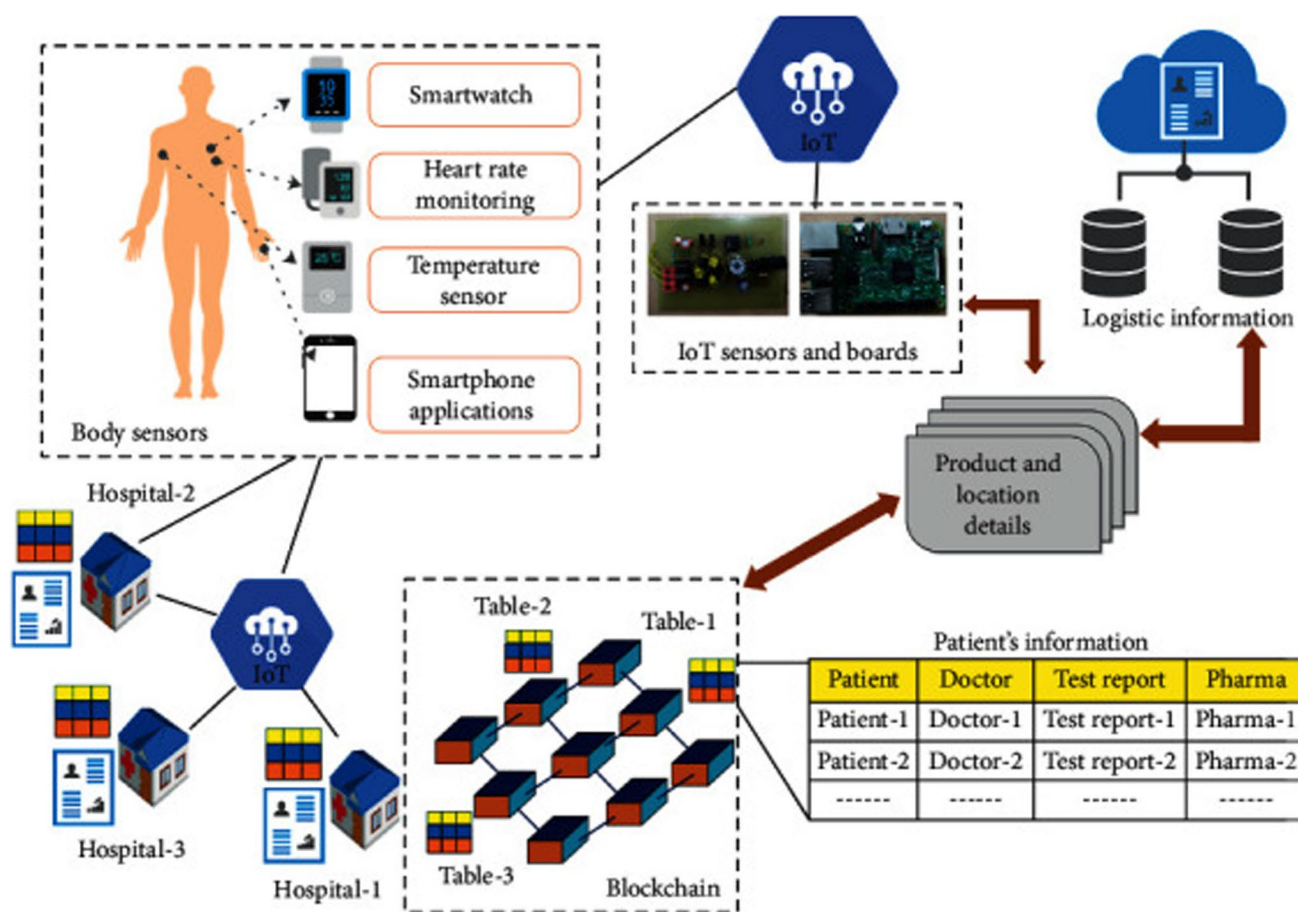


Fig. 5 Blockchain with Healthcare (Google Courtesy)

## 5 Healthcare IoT applications

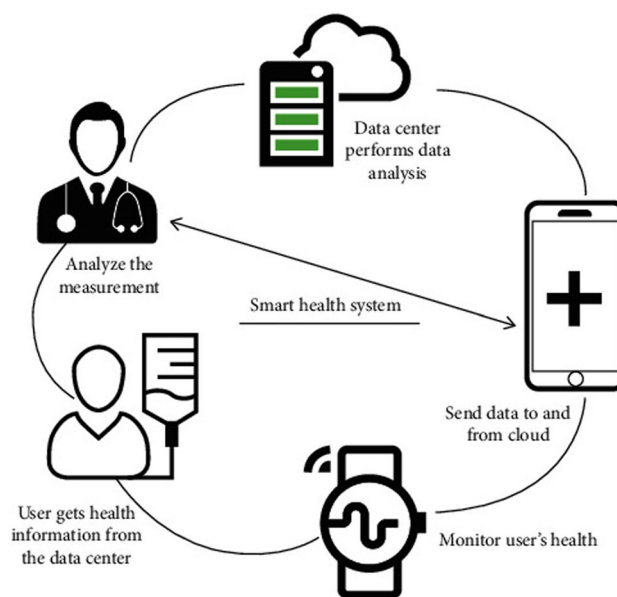
Many IoT applications have been built using HIoT services and principles. Experts in those domains have put forth many ideas for the benefit of humanity. To put it another way, applications focus on the end user, while concepts are more developer-centric. The rapid advancement of IoT-technology has resulted in the creation of wearable sensors, portable devices, and medical equipment that are both more affordable and easier to use. In the event of a medical emergency, these systems can gather patient data, diagnose illnesses, track patients' vitals, and send out alarms, as illustrated in Fig. 6. Figure 7 shows a variety of HIoT-based applications, some of which include visualising many states and others which only show one.

### 5.1 Monitorization of electrocardiogram

Heart electrical activity as a result of atrial and ventricular depolarization and repolarization can be captured by an electrocardiogram (ECG). A variety of cardiac problems can be detected with the help of an electrocardiogram (ECG), which records information on the fundamental rhythms of the heart muscles. Arrhythmia, myocardial ischemia, a prolonged QT interval, and other similar anomalies are examples of these. The IoT has shown promise in the field of electrocardiogram (ECG) monitoring for the early diagnosis of cardiac problems. IoT has been used in ECG monitoring by a number of previous research [39–44]. For real-time detection of cardiac abnormalities, it used a search automation method. A t-shirt was combined with a tiny, wearable, low-power electrocardiogram monitoring system in [26]. A biopotential chip was utilized to get high-quality electrocardiogram data. Afterwards, the end-users received the recorded data over Bluetooth. It was possible to use a mobile app to see the recorded electrocardiogram data. The



**Fig. 6** Patient data gathering using IoT (Google Courtesy)



**Fig. 7** HIoT based Applications



purpose of this system is to continuously evaluate the accelerometer and electrocardiogram data of older patients in order to give them real-time monitoring.

## 5.2 Tracking blood sugar levels

A high blood glucose level that does not go down for an extended length of time is the hallmark of diabetes. It ranks high among the illnesses that affect people. Type I diabetes, type 2 diabetes, and gestational diabetes are the three main forms of the disease. Oral glucose tolerance testing, fasting plasma glucose testing, and random plasma glucose testing can all help diagnose the illness and its subtypes. But "fingerpicking" and subsequent blood glucose level measurements are the gold standards for diabetes diagnosis. New Internet of Things (IoT) devices have allowed for the creation of a variety of noninvasive, comfortable, convenient, and safe wearable glucose monitors [45–48]. For continuous monitoring of blood glucose levels, a non-invasive glucometer based on mobile Internet of Things (m-IoT) was suggested in [28]. Here, IPv6 connection was used to link the healthcare practitioners with the wearable sensors. With the use of a Raspberry Pi

camera and a visible laser beam, Alarcón-Paredes et al. have developed a glove that can measure blood glucose levels. In order to diagnose diabetes, a series of images captured from the patient's fingertip were employed [29].

### 5.3 Tracking the temperature

Taking a patient's temperature is a crucial component of many diagnostic procedures since it shows how well the body is able to maintain homeostasis. Furthermore, in certain medical conditions like trauma, sepsis, etc., a rise or fall in core temperature might serve as an early warning signal. Medical professionals can learn a lot about a patient's health status from monitoring their temperature changes over time. Traditional methods of taking body temperatures involve placing a thermometer on the patient's ear, mouth, or rectum. However, these approaches always come with the drawbacks of low patient comfort and increased infection risks. New IoT technologies, however, have offered a number of potential answers to this issue. An earpiece that monitors the wearer's internal temperature via the tympanic membrane and an infrared sensor was suggested in [49] as a possible wearable medical gadget. A data processing unit and wireless sensor module were both integrated into the gadget. In this case, environmental factors and other physical activity have no effect on the observed temperature.

### 5.4 Monitoring blood pressure

Taking a patient's blood pressure is an essential part of any diagnostic procedure. At least one person is needed to record the blood pressure using the most common method. Nonetheless, blood pressure monitoring has undergone a sea change due to the incorporation of IoT and other sensing technologies. An example of a wearable cuffless device that can measure systolic and diastolic blood pressure is suggested in [50]. It is possible to save the recorded data in the cloud. In addition, 60 individuals were examined to determine the device's effectiveness and accuracy. The data might be saved by the device for later use as well. In this case, the blood pressure was determined with the help of the associated microcontroller module, and the resulting data was uploaded to a cloud storage service.

### 5.5 Monitoring of oxygen saturation

An important metric in healthcare analysis, pulse oximetry is a non-invasive way to assess oxygen saturation. The conventional strategy has its flaws, but the non-invasive method gives you real-time monitoring and gets rid of them. Improved pulse oximetry made possible by incorporating Internet of Things-based technologies has demonstrated promising use in healthcare. The blood oxygen saturation level, heart rate, and pulse parameters are measured by a noninvasive tissue oximeter in [51]. Zigbee and Wi-Fi are just two of the many possible communication technologies that could send the captured data to the server. The choice to intervene medically was based on the data that was recorded. This remote patient monitoring device is both economical and energy efficient. You may use the device for real-time monitoring with ease.

### 5.6 Monitoring asthma

Breathing becomes more difficult for people with asthma because of the chronic disease's impact on the airways. Asthma causes air passage swelling, which causes the airways to constrict. Wheezing, coughing, chest discomfort, and difficulty breathing are among the many symptoms that follow. Asthma attacks can strike at any time, and the only thing that can save you in that situation is an inhaler or nebulizer. As a result, keeping an eye on this condition in real-time could be necessary. There have been several proposals for Internet of Things (IoT) systems that monitor asthma in the past few years [52–54]. Asthma sufferers could keep track of their breathing rates with the help of a smart sensor in an HIoT system described in [54]. A cloud server houses the patient's medical records, which may be accessed by healthcare providers for the purposes of diagnosis and monitoring. Plus, it could assess the surrounding environment and tell the patient to leave if it wasn't healthy for him. Potentially necessary elements for an IoT asthma monitoring system in the future include the ability to track peak flows, pollen, humidity, ambient temperature, and asthma symptoms, among others.

## 5.7 Tracking emotions

In order to keep one's mind in good shape, mood tracking is useful since it reveals important details about one's emotional condition. Additionally, it aids medical personnel in their treatment of a wide range of mental illnesses, including but not limited to: bipolar disorder, depression, stress, and anxiety. By keeping track of one's emotions, one can gain a better grasp of their mental health. A CNN network was used to assess and classify an individual's emotional state into six distinct states: excited, furious, sad, calm, troubled, and excited (as reported in [55]). Integrating a state-of-the-art machine learning algorithm has made it possible to use heart rate as a predictor of impending stress. Another feature is that the system can relay information regarding the patient's stress level to them [57]. An intriguing side effect of stress state analysis is its potential utility in developing an Internet of Things (IoT) system for accident prevention.

## 5.8 Managing medications

In healthcare, medication adherence is a prevalent problem. Patients are at increased risk for unfavorable health problems if they do not follow to their drug schedule. The majority of cases of medication non-adherence occur among the elderly, who are more likely to experience cognitive decline, dementia, and other age-related clinical problems. As a result, they have a hard time carefully adhering to doctors' orders. Medication adherence tracking using the Internet of Things has been the subject of a great deal of prior research [56–59]. A smart medical box that can remind people to take their medication was invented in [60]. Every morning, afternoon, and night, there are three separate trays in the box that hold the medication. A number of critical health indicators (temperature, blood oxygen levels, glucose levels, electrocardiogram, etc.) are also measured by the device. The next step is to upload all of the captured data to a server in the cloud. The two end-users were able to communicate with each other through a mobile app. Using the mobile app, both the doctor and the patient can access the recorded information. In addition, a smart pharmaceutical system that is adaptive and built on the IoT [61] these system analyses temperature data using fuzzy logic. Continuous temperature monitoring allows the system to automatically modify the timing and dosage of medication during therapy, making it an effective tool for fever treatment.

## 5.9 Managing wheelchairs

For those with mobility impairments, a wheelchair is an essential component of their daily lives. Both physically and mentally, it helps them. But when a person's impairment is the result of brain damage, the wheelchair won't be able to do anything. Therefore, there is a renewed effort to find a way to link these wheelchairs to a navigation and tracking system. Systems built on the Internet of Things are now demonstrating promise in accomplishing this objective [62, 63]. Along with a real-time obstacle avoidance system, an IoT-based steering system has been suggested in [64]. The steering system is able to identify potential hazards by analyzing the recorded films in real-time using image processing algorithms. Wheelchair management has become more user-friendly and interactive thanks to mobile computing. Smart wheelchairs, like the one seen in [65], are the result of a combination of sensor technology, mobile computing, and the cloud. Patients and cares can communicate through a smartphone app that is part of the system. The software also lets cares keep an eye on the wheelchair even when they're not around. It should be noted that in [66], a more sophisticated and automated smart wheelchair was described that could detect obstacles, provide an umbrella, a foot mat, a head mat, and track the wheelchair's movement. More effective contact with the home environment was a result of the planned system in this case.

## 5.10 Rehabilitation System

Restoring a patient's functional ability is a common goal of physical medicine and rehabilitation. Rehabilitation include determining the source of the problem and assisting patients in returning to a normal life. IoT applications in rehabilitation are varied and extensive, including cancer, sports injuries, stroke, and other physical disabilities [67–70]. With the use of a multimodal sensor, a smart walker rehabilitation system has been developed that can track a patient's gait and assess several movement metrics. Orientation angle, elevation, force, and other movement matrices were measured by the smart walker as the patient utilized it. Doctors were able to access these records and generate diagnostic findings

through a mobile app. A smart wearable wristband, a robotic hand, and a machine learning algorithm were also part of a stroke rehabilitation system [71]. In order to measure, pre-process, and send the bio potential signal, the armband was created utilizing a textile electrode that is based on the IoT and uses little power. Medical practitioners could utilize the data collected to foretell their patients' recoveries and design individualized rehabilitation plans.

### 5.11 Additional notable uses

There is a wide variety of uses for HIoT beyond the aforementioned tasks. The proliferation of HIoT applications is directly proportional to the exponential rise in technological capability. A number of previously unproven research domains are now making effective use of this technology, thanks to the widespread use of IoT devices. Some examples of this are the detection of haemoglobin, therapy of cancer, aberrant cellular growth, remote surgery, etc. Several steps of cancer treatment, such as radiation and chemotherapy, were included into a new IoT architecture in [72]. Online consultations with doctors were conducted through a smartphone app. The healthcare practitioner had access to the patients' lab-test results saved in the cloud, which allowed them to determine the timing and amount of medication. Using a variety of cutting-edge machine learning algorithms in conjunction with an Internet of Things-based system to identify lung cancer is another possible use case [73–75]. Another way the device's efficacy was confirmed was by comparing the outcomes to the standard colorimetric test.

## 6 General case study

### 6.1 Remote patient monitoring (RPM)

- **Wearable sensors:** These are things like fitness trackers and smartwatches that can check your blood sugar, heart rate, and exercise level.
- **Mobile health applications:** These are apps that get information from smart tech and send it to doctors.
- **Cloud storage:** Safe cloud servers that keep patient data so that doctors can view it and look it over at any time.
- **Use case:** People who have long-term illnesses like diabetes or heart disease use personal sensors to keep an eye on their vital signs all the time. Mobile apps send data to healthcare providers, so they can keep an eye on things in real time and act quickly without having to make a lot of in-person trips.

### 6.2 Smart medication management

- **Smart pill bottles:** These are containers that have sensors built in to keep track of when medicine is taken.
- **Mobile health apps:** these are apps that tell patients to take their medicine and keep track of how well they do.
- **Cloud storage:** a safe place to store data about how well people are taking their medications that healthcare workers can access.
- **Use case:** People who take a lot of different medications use smart pill bottles that send alerts and keep track of when pills are taken. It is used to make sure that people take their medicine as prescribed and lower the risk of problems that can happen from missing doses.

### 6.3 Hospital asset tracking

- **RFID Tags:** These are tags that are put on medical tools to track its location in real time.
- **IoT Gateways:** These are machines that get information from RFID tags and send it to the main system in the hospital.
- **Central Management System:** This is software that processes all tagged equipment and shows where it is.
- **Use case:** Hospitals use RFID tags and Internet of Things (IoT) connections to keep track of where important medical devices like ventilators and infusion pumps are. This method makes sure that the right tools are available when they are needed, which cuts down on the time spent looking for them and boosts overall productivity.



## 7 Challenges and limitations associated with implementation of IoT in healthcare

### 7.1 Concerns about safety and privacy

- **Data breaches:** A lot of private patient data is created by IoT devices used in healthcare. Cybercriminals are very interested in this data, which makes IoT devices easy to hack and leak data from.
- **Unauthorised access:** It is very important to make sure that only authorised people can get to patient information. Weak security measures can allow people who aren't supposed to be there to get in, which can compromise patient privacy.
- **Encryption and authentication:** To protect the accuracy and privacy of data, it is important to use strong encryption and authentication methods. However, this can be hard to do properly and require a lot of resources.

### 7.2 Problems with Interoperability

- **Standards and procedures:** IoT devices may not work with each other properly if there aren't any standardised procedures for them. Devices from different companies might not be able to talk to each other properly.
- **Data integration:** It is very hard to combine data from many IoT devices into a code that makes sense and can be used. To do this, we need advanced standards for interoperability and data management tools.

### 7.3 Problems with scalability

- **Infrastructure:** To make IoT solutions work on a larger scale in healthcare, a lot of money needs to be spent on network bandwidth, storage, and computer power.
- **Device management:** It can be hard and take a lot of time and resources to keep track of a lot of IoT devices and make sure they get updates and are maintained throughout their lives.

### 7.4 Problems with rules and regulations

- **Rules for healthcare:** IoT systems need to follow strict rules for healthcare, like HIPAA in the US or GDPR in Europe. Finding your way around these regulatory settings can be hard and take a lot of time.
- **Approval and certification:** medical IoT devices often need to be approved and certified by regulatory bodies. This process can take a long time and cost a lot of money.

### 7.5 Quality and dependability of data

- **Accuracy and precision:** It is very important to make sure that the data received by IoT devices is correct and reliable. Bad data can cause evaluations and treatment plans to be wrong.
- **Device malfunctions:** IoT devices often have technical problems that stop them from working properly. This can cause problems with healthcare services and could even hurt patients.

### 7.6 Thoughts on cost

- **Initial investment:** Putting IoT solutions into place can have high start-up costs, such as the cost of devices, equipment, and training.
- **Upkeep costs:** IoT in healthcare can be expensive in the long run because of the costs of ongoing upkeep and operations.

### 7.7 Concerns about ethics

- **Patient permission:** It is very important to get informed permission from patients before collecting and using their data. They need to know who has access to their information and how it is being used.

- **Who owns the data:** It is very important to make it clear who owns the data that IoT devices produce. This includes dealing with problems like sharing data and who owns it.

## 7.8 Problems with the technology

- **Battery life:** A lot of IoT devices run on batteries, and batteries that don't last long can make it hard to keep tracking and collecting data.
- **Connectivity:** IoT devices need to be able to connect to the internet reliably in order to work properly. In places with bad connectivity, IoT devices may not work as well as they should.

## 8 Conclusion

An expanding field of study in healthcare is the Internet of Things' (IoT) potential. With these advancements, healthcare systems can better anticipate patients' health problems, diagnose them, treat them, and keep tabs on them both in and out of the hospital. More and more traditional methods of providing health services will be supplemented or replaced by IoT as the use of technology-enabled health services grows, allowing health systems to provide more adaptable models of care. When it comes to the supply and use of IoT devices in healthcare, however, a transparent and strong code of practice for data management, privacy, secrecy, and cybersecurity is essential. Only then can IoT be effectively implemented. This study also details the existing healthcare services that have investigated the use of IoT-based technology. Incorporating these ideas, IoT-technology has aided medical practitioners in tracking and diagnosing a wide range of conditions, measuring a plethora of health data, and establishing diagnostic centers in outlying areas. Because of this, healthcare is now more focused on the individual patient rather than the facility. We have also covered the latest trends in HIoT applications and their diverse range of uses. In addition, we have detailed the problems that have arisen throughout the HIoT system's development, production, and implementation. In the following years, progress and research will be centered around these difficulties. By focusing on enhancing security, reducing costs, improving data management, developing regulatory frameworks, ensuring interoperability, and designing user-centric solutions, the healthcare industry can fully leverage IoMT to deliver improved patient care, outcomes, and overall healthcare efficiency.

**Acknowledgements** This research received no external funding. The APC will be funded by Digital Transformation Portfolio, Tshwane University of Technology, Staatsartillerie Rd, Pretoria West, Pretoria 0183, South Africa.

**Author contributions** G.G: Conceptualization, Writing-Original Draft, Writing-Review and Editing Investigation. T.S: Conceptualization, Writing-Original Draft, Writing-Review and Editing Investigation. A.D: Conceptualization, Writing-Original Draft, Writing-Review and Editing Investigation. H.R.G: Conceptualization, Writing-Original Draft, Writing-Review and Editing Investigation. R.S: Conceptualization, Writing-Review and Editing, Supervision. A.G: Writing-Review and Editing, Supervision. L.R.G: Writing-Review and Editing, Supervision. A.K.T.: Writing-Review and Editing, Supervision; N.P.: Writing-Review and Editing, Supervision. B.T.: Writing-Review and Editing, Funding Acquisition, Supervision. All authors have read and agreed to the published version of the manuscript.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Ethics approval and consent to participate** Not applicable.

**Competing interests** The authors declare no competing interests.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

## References

1. Australia's Health. 2014. <http://www.aihw.gov.au/WorkArea/DownloadAsset.aspx?id=60129548150>. Accessed 08 Mar 2024.
2. Gope P, Hwang T. BSN-care: a secure IoT-based modern healthcare system using body sensor network. *IEEE Sensors J*. 2016;16:1368–76.
3. Fan YJ, Yin YH, Xu LD, Zeng Y, Wu F. IoT-based smart rehabilitation system. *IEEE Trans Ind Informat*. 2014;10(2):1568–77.
4. Ahmed A, Hassan I, El-Kady MF, Radhi A, Jeong CK, Selvaganapathy PR, Zu J, Ren S, Wang Q, Kaner RB. Integrated triboelectric nanogenerators in the era of the Internet of Things. *Adv Sci*. 2019;6:1802230.
5. Dobre C, Xhafa F. Intelligent services for big data science. *Future Gener Comput Syst*. 2014;37:267–81. <https://doi.org/10.1016/j.future.2013.07.014>.
6. Moosavi SR, Rahmani AM, Westerlund T, Yang G, Liljeberg P, Tenhunen H. Pervasive health monitoring based on Internet of Things: two case studies. In: 2014 4th International Conference on Wireless Mobile Communication and Healthcare-Transforming Healthcare Through Innovations in Mobile and Wireless Technologies (MOBIHEALTH). 2014. pp. 275–78.
7. Rahman M, Afsana F, Mahmud M, ShamimKaiser M, Ahmed MR, Kaiwartya O, James-Taylor A. Toward a heterogeneous mist, fog, and cloud-based framework for the internet of healthcare things. *IEEE Internet Things J*. 2019;6:4049–62. <https://doi.org/10.1109/JIOT.2018.2876088>.
8. Chirico A, Lucidi F, De Laurentiis M, Milanese C, Napoli A, Giordano A. Virtual reality in health system: beyond entertainment. A mini-review on the efficacy of VR during cancer treatment. *J Cell Physiol*. 2016;231(2):275–87.
9. de Carvalho M, Dias T, Duchesne M, Nardi A, Appolinario J. Virtual reality in health system: beyond entertainment. A mini-review on the efficacy of VR during cancer treatment. *Behav Sci (Basel)*. 2017;7(3):43.
10. Levine DM, Ouchi K, Blanchfield B, Diamond K, Licurse A, Pu CT, et al. Hospital-level care at home for acutely ill adults: a pilot randomized controlled trial. *J Gen Intern Med*. 2018;33(5):729–36.
11. Lu N, Li T, Ren X, Miao H. A deep learning scheme for motor imagery classification based on restricted boltzmann machines. *IEEE Trans Neural Syst Rehabil Eng*. 2017;25(6):566–76.
12. Mathews SM, Kambhamettu C, Barner KE. A novel application of deep learning for single-lead ECG classification. *Comput Biol Med*. 2018;1(99):53–62.
13. Australian Burden of Disease Study: Impact and Causes of Illness and Death in Australia 2015. Australian Institute of Health and Welfare. 2015. <https://www.aihw.gov.au/reports-data/health-conditions-disability-deaths/burden-of-disease/overview>. Accessed 02 Oct 2020.
14. de Silva D. Evidence: helping people help themselves. The Heart Foundation. 2011. <https://www.health.org.uk/publications/evidence-helping-people-help-themselves>. Accessed 02 Feb 2020.
15. Vickery DM, Kalmer H, Lowry D, Constantine M, Wright E, Loren W. Effect of a self-care education program on medical visits. *J Am Med Assoc*. 1983;250(21):2952–6.
16. Pai A. Survey: percent of smartphone users have downloaded a fitness or health app. *Mobile health news*. 2015. <https://www.mobihealthnews.com/48273/survey-58-percent-of-smartphone-users-have-downloaded-a-fitness-or-health-app>. Accessed 4 Apr 2020.
17. Nadarzynski T, Miles O, Cowie A, Ridge D. Acceptability of artificial intelligence (AI)-led chatbot services in healthcare: a mixed-methods study. *Digit Health*. 2019;5:2055207619871808.
18. Wisniewski H, Liu G, Henson P, Vaidyam A, Hajratalli NK, Onnela J, et al. Understanding the quality, effectiveness and attributes of top-rated smartphone health apps. *Evid Based Ment Health*. 2019;22(1):4–9.
19. Bates DW, Landman A, Levine DM. Health apps and health policy: what is needed? *J Am Med Assoc*. 2018;320(19):1975–6.
20. Borycki E. Quality and safety in ehealth: the need to build the evidence base. *J Med Internet Res*. 2019;21(12):e16689.
21. Marques G, Pitarna R. An indoor monitoring system for ambient assisted living based on internet of things architecture. *Int J Environ Res Public Health*. 2016;13(11):1152. <https://doi.org/10.3390/ijerph13111152>.
22. Bathilde JB. Continuous heart rate monitoring system as an IoT edge device. In: Proceedings of 2018 IEEE sensors applications symposium (SAS). Seoul; 2018. pp. 1–6.
23. Drew BJ, Califf RM, Funk M, et al. Practice standards for electrocardiographic monitoring in hospital settings. *Circulation*. 2004;110(17):2721–46. <https://doi.org/10.1161/01.cir.0000145144.56673.59>.
24. Dash P. Electrocardiogram monitoring. *Indian J Anaesth*. 2002;46:251–60.
25. Agu E. The smartphone as a medical device: assessing enablers, benefits and challenges. In: Proceedings of the 2013 IEEE International workshop of internet-of-things networking and control (IoT-NC). New Orleans; 2013. pp. 48–52.
26. Wu T. Advances in body area networks I. In: Fortino G, Wang Z, editors. A wearable, low-power, real-time ECG monitor for smart t-shirt and IoT healthcare applications. Berlin: Springer; 2019. p. 165–73.
27. Djelouat H, Al DM, Boukhenoufa I, et al. Real-time ECG monitoring using compressive sensing on a heterogeneous multicore edge-device. *Microprocess Microsyst*. 2020;72: 102839. <https://doi.org/10.1016/j.micpro.2019.06.009>.
28. Istepanian RS. The potential of Internet of m-health Things “m-IoT” for non-invasive glucose level sensing. In: Proceedings of the 2011 Annual international conference of the IEEE engineering in medicine and biology society. Boston; 2011. pp. 5264–5266.
29. Alarcón-Paredes A, Francisco-García V, Guzmán-Guzmán IP, Cantillo-Negrete J, Cuevas-Valencia RE, Alonso-Silverio GA. An IoT-based non-invasive glucose level monitoring system using raspberry pi. *Appl Sci*. 2019;9(15):3046. <https://doi.org/10.3390/app9153046>.
30. Dohr A. The internet of things for ambient assisted living. In: Proceedings of the 2010 Seventh international conference on information technology: new generations. Las Vegas; 2010. pp. 804–809.
31. Tsirmpas C, Anastasiou A, Bountris P, Koutsouris D. A new method for profile generation in an internet of things environment: an application in ambient-assisted living. *IEEE Internet Things J*. 2015;2(6):471–8. <https://doi.org/10.1109/jiot.2015.2428307>.
32. Maskeliūnas R. A review of internet of things technologies for ambient assisted living environments. *Future Internet*. 2019;11:259.
33. Tabish R. A 3G/WiFi-enabled 6LoWPAN-based U-healthcare system for ubiquitous real-time monitoring and data logging. In: Proceedings of the 2nd Middle East conference on biomedical engineering. Doha; 2014. pp. 277–280.
34. Zhang Y, Cui J, Ma K, Chen H, Zhang J. A wristband device for detecting human pulse and motion based on the internet of things. *Measurement*. 2020. <https://doi.org/10.1016/j.measurement.2020.108036>.

35. Singh K. Proceedings of the Third international conference on computational intelligence and informatics. In: Raju KS, Govardhan A, Rani BP, Sridevi R, Murty MR, editors. Role and impact of wearables in IoT healthcare. Berlin: Springer; 2020. p. 735–42.
36. Wang W. The internet of things for resident health information service platform research. In: Proceedings of the IET International conference on communication technology and application (ICCTA 2011). Beijing; 2011.
37. Behera RK, Bala PK, Dhir A. The emerging role of cognitive computing in healthcare: a systematic literature review. *Int J Med Informatics*. 2019;129:154–66. <https://doi.org/10.1016/j.ijmedinf.2019.04.024>.
38. Amin SU, Hossain MS, Muhammad G, Alhussein M, Rahman MA. Cognitive smart healthcare for pathology detection and monitoring. *IEEE Access*. 2019;7:10745–53. <https://doi.org/10.1109/access.2019.2891390>.
39. Jara AJ. A pharmaceutical intelligent information system to detect allergies and adverse drugs reactions based on internet of things. In: Proceedings of the 2010 8th IEEE international conference on pervasive computing and communications workshops (PERCOM Workshops). Mannheim; 2010. pp. 809–812.
40. Satamraju KP, Malarkodi B. Proof of concept of scalable integration of internet of things and blockchain in healthcare. *Sensors*. 2020;20(5):1389. <https://doi.org/10.3390/s20051389>.
41. Zhang X., Poslad S. Blockchain support for flexible queries with granular access control to electronic medical records (EMR). In: Proceedings of the 2018 IEEE International conference on communications (ICC). Kansas City; 2018. pp. 1–6.
42. Yue X, Wang H, Jin D, Li M, Jiang W. Healthcare data gateways: found healthcare intelligence on blockchain with novel privacy risk control. *J Med Syst*. 2016;40:218. <https://doi.org/10.1007/s10916-016-0574-6>.
43. Nigar N., Chowdhury L. An intelligent children healthcare system by using ensemble technique. In: Proceedings of international joint conference on computational intelligence. Budapest; pp. 137–150.
44. Sutjiredjeki E, Basjaruddin NC, Fajrin DN, Noor F. Development of NFC and IoT-enabled measurement devices for improving health care delivery of Indonesian children. *Journal of Phys Conf Ser*. 2020;1450: 012072.
45. Vazquez-Briseno M. A proposal for using the internet of things concept to increase children's health awareness. In: Proceedings of CONIELECOMP 2012, 22nd International conference on electrical communications and computers. Cholulu; 2012. pp. 168–72.
46. Tekeste T. Ultra-low power QRS detection and ECG compression architecture for IoT healthcare devices. *IEEE Trans Circuits Syst I Regul Pap*. 2018;66:669–79.
47. Liu M-L. Internet of Things-based electrocardiogram monitoring system. *Chin Patent*. 2012;102:118.
48. Gia TN, et al. Energy efficient fog-assisted IoT system for monitoring diabetic patients with cardiovascular disease. *Futur Gener Comput Syst*. 2019;93:198–211.
49. Bhat G. M., Bhat N. G. A novel IoT based framework for blood glucose examination. In: Proceedings of the 2017 International conference on electrical, electronics, communication, computer, and optimization techniques (ICECCOT). Mysuru; 2017. pp. 205–7.
50. Gia TN, Ali M, Dhaou IB, et al. IoT-based continuous glucose monitoring system: a feasibility study. *Proc Comp Sci*. 2017;109:327–34. <https://doi.org/10.1016/j.procs.2017.05.359>.
51. Sargunam B., Anusha S. IoT based mobile medical application for smart insulin regulation. In: Proceedings of the 2019 IEEE International conference on electrical, computer and communication technologies (ICECT). Erode; 2019. pp. 1–5.
52. Ota H, Chao M, Gao Y, et al. 3D printed “earable” smart devices for real-time detection of core body temperature. *ACS Sensors*. 2017;2(7):990–7. <https://doi.org/10.1021/acssensors.7b00247>.
53. Xin Q, Wu J. A novel wearable device for continuous, non-invasion blood pressure measurement. *Comput Biol Chem*. 2017;69:134–7. <https://doi.org/10.1016/j.combiolchem.2017.04.011>.
54. Fu Y, Liu J. System design for wearable blood oxygen saturation and pulse measurement device. *Proc Manuf*. 2015;3:1187–94. <https://doi.org/10.1016/j.promfg.2015.07.197>.
55. Güler İ. Real-time abnormal detection for asthma patients with internet of things technology. In: Proceedings of the 2018 3rd International conference on computer science and engineering (UBMK). Sarajevo, Bosnia and Herzegovina; 2018. pp. 269–274.
56. Li B, Dong Q, Downen RS, et al. A wearable IoT aldehyde sensor for pediatric asthma research and management. *Sens Actuators B Chem*. 2019;287:584–94. <https://doi.org/10.1016/j.snb.2019.02.077>.
57. Gurbeta L, Badnjevic M, Maksimovic M, Omanovic-Miklicanin E, Sejdic E. A telehealth system for automated diagnosis of asthma and chronic obstructive pulmonary disease. *J Am Med Inform Assoc*. 2018;25(9):1213–7. <https://doi.org/10.1093/jamia/ocy055>.
58. Alam MG. CNN based mood mining through IoT-based physiological sensors observation. *Korea Inform Sci Soc*. 2017. pp. 1301–1303.
59. Aldeer M, Javanmard M, Martin R. A review of medication adherence monitoring technologies. *Appl Syst Innov*. 2018;1(2):14. <https://doi.org/10.3390/asi1020014>.
60. Latif G. I-CARES: advancing health diagnosis and medication through IoT. *Wireless Netw*. 2019;4:1–15.
61. Sahlab N. Development of an intelligent pill dispenser based on an IoT-approach. In: Ahram T, Karwowski W, Pickl S, Taia R, editors. Proceedings of the international conference on human systems engineering and design future trends and applications. Munich: Springer; 2019. p. 33–9.
62. Shreyas A. Emerging research in computing, information, communication and applications. In: Shetty NR, Patnaik LM, Nagaraj HC, Hamsavath PN, Nalini N, editors. IoT-enabled medicine bottle. Berlin: Springer; 2019. p. 127–39.
63. Bharadwaj SA. Enhancing healthcare using m-care box (monitoring non-compliance of medication). In: Proceedings of the 2017 International conference on I-SMAC (IoT in social, mobile, analytics and cloud) (I-SMAC). Coimbatore; 2017. pp. 352–356.
64. Medina J, Espinilla M, García-Fernández ÁL, Martínez L. Intelligent multi-dose medication controller for fever: from wearable devices to remote dispensers. *Comput Electr Eng*. 2018;65:400–12. <https://doi.org/10.1016/j.compeleceng.2017.03.012>.
65. Ghorbel A, Amor NB, Jallouli M. A survey on different human-machine interactions used for controlling an electric wheelchair. *Proc Comp Sci*. 2019;159:398–407. <https://doi.org/10.1016/j.procs.2019.09.194>.
66. Ramya K, Nargees S, Tabasuum SA, Khan S, Shiny M. A survey on Smart automated WheelChair system with voice controller using IOT along with health monitoring for physically challenged persons. *Int Sci J Contemp Res Eng Sci Manag*. 2020;5:95–8.
67. Lee YK. Real time image processing based obstacle avoidance and navigation system for autonomous wheelchair application. In: Proceedings of the 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC). Kuala Lumpur; 2017. pp. 380–385.

68. Ghorbel A. Cloud based mobile application for remote control of intelligent wheelchair. In: Proceedings of the 2018 14th International wireless communications and mobile computing conference (IWCMC). Limassol; 2018. pp. 1249–1254.
69. Kumar D, Malhotra R, Sharma SR. Design and construction of a smart wheelchair. *Proc Comput Sci.* 2020;172:302–7. <https://doi.org/10.1016/j.procs.2020.05.048>.
70. Onasanya A, Elshakankiri M. Smart integrated IOT healthcare system for cancer care. *Wirel Netw.* 2019. <https://doi.org/10.1007/s11276-018-01932-1>.
71. Adamovich SV, Merians AS, Boian R, et al. A virtual reality-based exercise system for hand rehabilitation post-stroke. *Presence Teleoperators Virtual Environ.* 2005;14(2):161–74. <https://doi.org/10.1162/1054746053966996>.
72. Qi J, Yang P, Waraich A, Deng Z, Zhao Y, Yang Y. Examining sensor-based physical activity recognition and monitoring for healthcare using internet of things: a systematic review. *J Biomed Inform.* 2018;87:138–53. <https://doi.org/10.1016/j.jbi.2018.09.002>.
73. Yang G, Deng J, Pang G, et al. An IoT-enabled stroke rehabilitation system based on smart wearable armband and machine learning. *IEEE J Transl Eng Health Med.* 2018;6:1–10. <https://doi.org/10.1109/jtehm.2018.2822681>.
74. Heshmat M, Shehata ARS. A framework about using internet of things for smart cancer treatment process. In: Proceedings of the International conference on industrial engineering and operations management. Washington. 2018. pp. 1206–1211.
75. Pradhan K, Chawla P. Medical Internet of things using machine learning algorithms for lung cancer detection. *J Manag Anal.* 2020;7(4):591–623. <https://doi.org/10.1080/23270012.2020.1811789>.

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.